

COER UNIVERSITY, ROORKEE
Department of Computer Science & Engineering
Full Stack Development — Mini E-Commerce Project

Lab Sheet 2: Making It Interactive (Vanilla JavaScript)
Covers Project Phase 3

Name:- Kashish Chauhan

Btech(cse) 3rd year 5th sem

Sec: B , Rollno: 28

CUID: cu240251660

[script.js](#)

```
const products = [
  {
    id: 1,
    name: "Wireless Headphones",
    price: 1999,
    image:
      "https://images.unsplash.com/photo-1505740420928-5e560c06d30e?auto=format&fit=crop&w=600&q=80"
  },
  {
    id: 2,
    name: "Smart Watch",
    price: 2499,
    image:
      "https://images.unsplash.com/photo-1523275335684-37898b6baf30?auto=format&fit=crop&w=600&q=80"
  },
  {
    id: 3,
    name: "Running Shoes",
    price: 1799,
    image:
      "https://images.unsplash.com/photo-1542291026-7eec264c27ff?auto=format&fit=crop&w=600&q=80"
  },
  {
    id: 4,
    name: "Travel Backpack",
```

```

        price: 1299,
        image:
"https://images.unsplash.com/photo-1553062407-98eeb64c6a62?auto=format&fit=cr
op&w=600&q=80"
    },
    {
        id: 5,
        name: "Bluetooth Speaker",
        price: 1499,
        image:
"https://images.unsplash.com/photo-1608043152269-423dbba4e7e1?auto=format&fi
t=crop&w=600&q=80"
    },
    {
        id: 6,
        name: "Laptop Stand",
        price: 899,
        image: "https://loremflickr.com/600/400/laptop,stand"
    },
    {
        id: 7,
        name: "USB-C Hub",
        price: 1099,
        image: "https://loremflickr.com/600/400/usb,hub"
    },
    {
        id: 8,
        name: "Mechanical Keyboard",
        price: 2999,
        image:
"https://images.unsplash.com/photo-1587829741301-dc798b83add3?auto=format&fi
t=crop&w=600&q=80"
    }
];

```

```

function getCart() {
    return JSON.parse(localStorage.getItem("cart")) || [];
}

```

```

function saveCart(cart) {
    localStorage.setItem("cart", JSON.stringify(cart));
}

```

```

function updateCartCount() {

```

```

const cart = getCart();
const count = cart.reduce((total, item) => total + item.quantity, 0);

document.querySelectorAll(".cart-count").forEach(element => {
  element.textContent = count;
});
}

function displayProducts() {
  const productGrid = document.querySelector(".product-grid");

  if (!productGrid) return;

  productGrid.innerHTML = "";

  products.forEach(product => {
    const card = document.createElement("div");
    card.className = "product-card";

    card.innerHTML = `
      
      <h3>${product.name}</h3>
      <p class="price">₹${product.price.toLocaleString("en-IN")}</p>
      <button class="add-cart-button" data-id="${product.id}">
        Add to Cart
      </button>
    `;

    productGrid.appendChild(card);
  });

  document.querySelectorAll(".add-cart-button").forEach(button => {
    button.addEventListener("click", () => {
      addToCart(Number(button.dataset.id));
    });
  });
}

function addToCart(productId) {
  const cart = getCart();
  const product = products.find(item => item.id === productId);

  if (!product) return;

```

```

const existingItem = cart.find(item => item.id === productId);

if (existingItem) {
  existingItem.quantity++;
} else {
  cart.push({
    id: product.id,
    name: product.name,
    price: product.price,
    image: product.image,
    quantity: 1
  });
}

saveCart(cart);
updateCartCount();
}

function displayCart() {
  const cartBody = document.getElementById("cart-body");

  if (!cartBody) return;

  const cart = getCart();
  cartBody.innerHTML = "";

  if (cart.length === 0) {
    cartBody.innerHTML = `
      <tr>
        <td colspan="5" class="empty-cart">
          Your cart is empty.
        </td>
      </tr>
    `;

    calculateTotal();
    return;
  }

  cart.forEach(item => {
    const row = document.createElement("tr");

    row.innerHTML = `
      <td>

```

```

        
    </td>

    <td>\${item.name}</td>

    <td>
        <input
            type="number"
            min="1"
            value="\${item.quantity}"
            class="quantity-input"
            data-id="\${item.id}"
        >
    </td>

    <td>₹\${item.price.toLocaleString("en-IN")}</td>

    <td>
        ₹\${(item.price * item.quantity).toLocaleString("en-IN")}
        <br>
        <button class="remove-button" data-id="\${item.id}">
            Remove
        </button>
    </td>
`
;

cartBody.appendChild(row);
});

document.querySelectorAll(".quantity-input").forEach(input => {
    input.addEventListener("change", () => {
        updateQuantity(
            Number(input.dataset.id),
            Number(input.value)
        );
    });
});

document.querySelectorAll(".remove-button").forEach(button => {
    button.addEventListener("click", () => {
        removeFromCart(Number(button.dataset.id));
    });
});

```

```

    calculateTotal();
}

function updateQuantity(productId, quantity) {
    const cart = getCart();
    const item = cart.find(product => product.id === productId);

    if (!item) return;

    item.quantity = quantity < 1 || isNaN(quantity) ? 1 : quantity;

    saveCart(cart);
    displayCart();
    updateCartCount();
}

function removeFromCart(productId) {
    const cart = getCart().filter(item => item.id !== productId);

    saveCart(cart);
    displayCart();
    updateCartCount();
}

function calculateTotal() {
    const totalElement = document.getElementById("grand-total");

    if (!totalElement) return;

    const cart = getCart();

    const total = cart.reduce(
        (sum, item) => sum + item.price * item.quantity,
        0
    );

    totalElement.textContent = `₹${total.toLocaleString("en-IN")}`;
}

function validateCheckout(event) {
    event.preventDefault();

    const name = document.getElementById("name");
    const address = document.getElementById("address");

```

```
const city = document.getElementById("city");
const pincode = document.getElementById("pincode");
const phone = document.getElementById("phone");

const nameError = document.getElementById("name-error");
const addressError = document.getElementById("address-error");
const cityError = document.getElementById("city-error");
const pincodeError = document.getElementById("pincode-error");
const phoneError = document.getElementById("phone-error");
const paymentError = document.getElementById("payment-error");

document.querySelectorAll(".error").forEach(error => {
  error.textContent = "";
});

let valid = true;

if (name.value.trim() === "") {
  nameError.textContent = "Name is required.";
  valid = false;
}

if (address.value.trim() === "") {
  addressError.textContent = "Address is required.";
  valid = false;
}

if (city.value.trim() === "") {
  cityError.textContent = "City is required.";
  valid = false;
}

if (!/^\d{6}$/.test(pincode.value.trim())) {
  pincodeError.textContent = "Pincode must be exactly 6 digits.";
  valid = false;
}

if (!/^\d{10}$/.test(phone.value.trim())) {
  phoneError.textContent = "Phone must be exactly 10 digits.";
  valid = false;
}

const payment = document.querySelector(
  'input[name="payment"]:checked'
```

```

);

if (!payment) {
    paymentError.textContent = "Please select a payment method.";
    valid = false;
}

if (!valid) return;

localStorage.removeItem("cart");
updateCartCount();

document.querySelector(".checkout-form").style.display = "none";

document.getElementById("order-message").innerHTML = `
    <div class="order-success">
        <h2>Order placed!</h2>
        <p>Your order has been placed successfully.</p>
        <a href="index.html" class="checkout-button">
            Continue Shopping
        </a>
    </div>
`;
}

document.addEventListener("DOMContentLoaded", () => {
    displayProducts();
    displayCart();
    updateCartCount();

    const checkoutForm = document.querySelector(".checkout-form");

    if (checkoutForm) {
        checkoutForm.addEventListener("submit", validateCheckout);
    }
});

```

Output:-

